

From Scaffolding to Governance: The Future of AI Work

Reuben Brasher

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Lecture 9

From Tools to Systems

AI development progresses from local scaffolding to organizational governance.

- ▶ Prompting introduces initial constraints.
- ▶ Workflows structure interactions.
- ▶ Systems coordinate components.
- ▶ Governance formalizes constraints.
- ▶ Usage evolves toward structure.
- ▶ Control scales institutionally.

A Single Trajectory

The series describes a progression of increasing constraint.

- ▶ Phase one focused on cognition.
- ▶ Phase two introduced workflows.
- ▶ Phase three revealed failures.
- ▶ Phase four introduced governance.
- ▶ Each phase added control.
- ▶ Structure increased progressively.

Core Synthesis

Constraint is required to stabilize inherently unconstrained systems.

- ▶ LLMs produce unconstrained outputs.
- ▶ Structure limits output space.
- ▶ Constraints improve reliability.
- ▶ Lack of constraints causes failure.
- ▶ Control must be imposed.
- ▶ Stability emerges from layering.

Scaffolding Defined

Scaffolding enables reliable interaction with models.

- ▶ It guides model behavior.
- ▶ It reduces ambiguity.
- ▶ It aligns representations.
- ▶ It enables refinement.
- ▶ It supports validation.
- ▶ It acts as control.

Forms of Scaffolding

Scaffolding appears in multiple forms.

- ▶ Priming establishes context.
- ▶ Schema defines structure.
- ▶ Decomposition breaks tasks.
- ▶ Testing verifies correctness.
- ▶ Role separation isolates work.
- ▶ Each form constrains failure.

Reliability Through Scaffolding

Reliability depends on structured scaffolding.

- ▶ Unstructured prompts are unstable.
- ▶ Workflows improve consistency.
- ▶ Validation prevents errors.
- ▶ Decomposition reduces complexity.
- ▶ Role separation limits propagation.
- ▶ Structure stabilizes outputs.

Critical Shift

Scaffolding is the system rather than a workaround.

- ▶ It defines model usage.
- ▶ It determines output quality.
- ▶ It replaces internal reasoning.
- ▶ It creates reproducibility.
- ▶ It enables collaboration.
- ▶ It transforms tools into systems.

From Prompts to Workflows

Prompts evolve into structured workflows.

- ▶ Prompts become multi-step processes.
- ▶ Steps are sequenced.
- ▶ Outputs feed forward.
- ▶ Validation is inserted.
- ▶ Roles are assigned.
- ▶ Workflows become repeatable.

From Workflows to Pipelines

Workflows scale into pipelines.

- ▶ Processes become standardized.
- ▶ Steps are automated.
- ▶ Data flows are explicit.
- ▶ Dependencies are managed.
- ▶ Outputs are structured.
- ▶ Pipelines enable scale.

From Pipelines to Systems

Pipelines evolve into integrated systems.

- ▶ Multiple pipelines interact.
- ▶ Data flows across components.
- ▶ Systems integrate infrastructure.
- ▶ Monitoring is introduced.
- ▶ Control is formalized.
- ▶ Systems operate continuously.

Key Insight

Prompting is an early form of system design.

- ▶ Prompts define inputs.
- ▶ Workflows define logic.
- ▶ Validation defines correctness.
- ▶ Roles define architecture.
- ▶ Iteration defines development.
- ▶ Design emerges from interaction.

Transition to Governance

Systems that process data must be governed.

- ▶ Data flows require control.
- ▶ Outputs require accountability.
- ▶ Processes require documentation.
- ▶ Systems must meet regulation.
- ▶ Governance ensures consistency.
- ▶ Control scales organizationally.

Scaffolding to Governance

Local scaffolding maps to governance structures.

- ▶ Prompts become policies.
- ▶ Schema becomes standards.
- ▶ Testing becomes auditing.
- ▶ Roles become organizational roles.
- ▶ Validation becomes compliance.
- ▶ Workflows become governed processes.

Governance as Scaffolding

Governance is system-level scaffolding.

- ▶ Defines system behavior.
- ▶ Enforces constraints.
- ▶ Standardizes practices.
- ▶ Ensures compliance.
- ▶ Enables accountability.
- ▶ Sustains reliability.

Reinterpreting Agentic AI

Agentic AI is structured and governed workflow execution.

- ▶ Agents execute processes.
- ▶ Memory provides state.
- ▶ Constraints define actions.
- ▶ Outputs are pipeline artifacts.
- ▶ Behavior must be controlled.
- ▶ Autonomy is bounded.

Core Definition of an Agent

An agent is a constrained pipeline with memory.

- ▶ It receives defined inputs.
- ▶ It processes structured steps.
- ▶ It maintains state.
- ▶ It produces constrained outputs.
- ▶ It operates under governance.
- ▶ It is not autonomous.

Hype vs Reality

Agent narratives obscure system realities.

- ▶ Hype emphasizes autonomy.
- ▶ Reality emphasizes control.
- ▶ Hype suggests independence.
- ▶ Reality requires supervision.
- ▶ Hype ignores governance.
- ▶ Reality depends on it.

Future Skill Shift

AI changes required competencies.

- ▶ Less focus on coding.
- ▶ Less focus on basic writing.
- ▶ More focus on structuring problems.
- ▶ More focus on workflow design.
- ▶ More focus on validation.
- ▶ More focus on systems thinking.

New Core Competencies

Expertise shifts toward design and supervision.

- ▶ Define clear boundaries.
- ▶ Design workflows.
- ▶ Evaluate outputs.
- ▶ Detect errors.
- ▶ Integrate systems.
- ▶ Maintain alignment.

Emerging Roles

New roles reflect structured interaction.

- ▶ AI architects design systems.
- ▶ Reviewers validate outputs.
- ▶ Governance specialists enforce policy.
- ▶ Engineers integrate systems.
- ▶ Compliance ensures alignment.
- ▶ Roles mirror workflows.

Human–AI Collaboration

LLMs act as junior collaborators.

- ▶ Models generate outputs.
- ▶ Humans define constraints.
- ▶ Humans validate correctness.
- ▶ Humans make decisions.
- ▶ Collaboration is iterative.
- ▶ Responsibility remains human.

Boundaries of Autonomy

AI autonomy remains limited.

- ▶ Systems operate within constraints.
- ▶ Outputs require validation.
- ▶ Decisions need oversight.
- ▶ Errors are externally detected.
- ▶ Governance defines limits.
- ▶ Autonomy is constrained.

Organizational Implications

AI systems must align with enterprise environments.

- ▶ Integration with infrastructure is required.
- ▶ Compliance must be maintained.
- ▶ Security must be ensured.
- ▶ Collaboration is necessary.
- ▶ Governance must be embedded.
- ▶ Systems must align with constraints.

Organizational Risk

Ignoring governance leads to failure.

- ▶ Compliance violations may occur.
- ▶ Security risks increase.
- ▶ Outputs may be unreliable.
- ▶ Stakeholders reject systems.
- ▶ Rework becomes necessary.
- ▶ Projects fail.

Strategic Positioning

Value lies in system design rather than models.

- ▶ Models are commoditized.
- ▶ Integration creates differentiation.
- ▶ Control ensures reliability.
- ▶ Governance enables adoption.
- ▶ Systems drive value.
- ▶ Strategy focuses on constraints.

Differentiation from Hype

Model-centric thinking fails in practice.

- ▶ Performance metrics are insufficient.
- ▶ Ignoring governance limits adoption.
- ▶ Autonomy narratives mislead.
- ▶ Systems thinking adds value.
- ▶ Integration matters more.
- ▶ Reliability outweighs capability.

Final Synthesis

Scaffolding evolves into governance systems.

- ▶ Starts with prompts.
- ▶ Expands into workflows.
- ▶ Scales into pipelines.
- ▶ Becomes governance.
- ▶ Enables reliability.
- ▶ Defines future systems.

Closing Statement

The future of AI is governed collaboration.

- ▶ Intelligence is insufficient.
- ▶ Structure enables reliability.
- ▶ Governance ensures safety.
- ▶ Collaboration defines effectiveness.
- ▶ Systems determine outcomes.
- ▶ The future is constrained.

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